Week 2 Tutorial: Stoichiometry and Gravimetric Analysis

Learning objectives:

At the end of this tutorial, you should be able to:

- calculate the amount of reactant/ product in a stoichiometric problem;
- identify limiting reactant in a problem involving two reactants;
- calculate percentage of yield in a reaction;
- solve problems related to gravimetric analysis

Key formulae:

1. To convert mole and mass of chemicals:

   $\boldsymbol{n = \frac{m}{M}}$ m = mass in gram, g
   M = molar mass in g mol^{-1}

2. To convert mole and volume of gaseous compounds:

   $\boldsymbol{n = \frac{PV}{RT}}$ (for gas only) P = pressure in kPa;
   V = volume of gas in dm^3 or litre
   R = gas constant, 8.314 J K^{-1} mol^{-1}
   T = temperature in Kelvin (T in ^{0}C + 273 = T in K)

   $\boldsymbol{n = \frac{V}{V_m}}$ (for gas only) V_m = molar volume of gas, L mol^{-1}
   At STP (0^{0}C and 101 kPa) V_m= 22.4 L mol^{-1}
   At SLR (25^{0}C and 101 kPa) V_m= 24.5 L mol^{-1}

3. To convert mole, concentration and volume of solutions:

   $\boldsymbol{n = CV}$ where C= concentration in mol L^{-1},
   V= volume in liter, L

4. To find percentage of yield (mass of product):

$$\%\ yield = \frac{actual\ mass\ of\ product}{theoretical\ mass\ of\ product} \times 100\%$$

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Worked Example:

Example 2.1

Lithium hydroxide reacts with the carbon dioxide according to this equation:

$$LiOH(s) + CO_2(g) \rightarrow LiHCO_3(aq)$$

What is the maximum volume of CO_2 that can be absorbed at SLC by 10 g of lithium hydroxide?

Step 1: Convert mass of known substance (10g LiOH) into number of mole

$$n\,(LiOH) = \frac{m}{M} = \frac{10\ g}{(6.9 + 16 + 1) g mol^{-1}} = 0.418\ mol$$

Step 2: Use mole ratio between known substance and unknown substance (i.e. CO_2) to find number of mole of unknown

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$$\frac{n(CO_2)}{n(LiOH)} = \frac{1}{1} \quad n(CO_2) = \frac{1}{1} \times 0.418\ mol = 0.418\ mol$$

Step 3: Convert the number of mole into desired quantities (i.e. volume at SLC),

$$n = \frac{V}{Vm} = \frac{V}{24.5\ L mol^{-1}} \quad V(CO_2) = 0.418\ mol\ \times 24.5\ L mol^{-1} = \mathbf{10.25\ L}$$

Example 2.2

In the following reaction, 6 mol of aluminium is mixed with 6 mol of oxygen:

$$4\ Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$$

What amount of aluminium oxide will be produced from this reaction?

Reaction	4 Al(s) +	3 O_2(g) →	$2Al_2O_3$(s)
Initial Amount	6 mol	6 mol	0 mol
Divided by Coefficient	$\frac{6}{4} = 1.5$	$\frac{6}{3} = 2$	
Identify As	Limiting Reactant	Reactant In Excess	
Amount Reacted	6 mol	$\frac{n\,(O_2)}{n(Al)} = \frac{3}{4}$ $n(O_2) = \frac{3}{4} \times 6\ mol$ $= 4.5\ mol$	$\frac{n\,(Al_2O_3)}{n(Al)} = \frac{2}{4}$ $n(Al_2O_3) = \frac{2}{4} \times 6\ mol$ $= 3.0\ mol$
Amount Left	0 mol	1.5 mol	3 mol

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Example 2.3:

What is the percent yield for the reaction between 9.8 grams of $Ca(OH)_2$ and 9.8 grams of H_3PO_4 when 2.5 grams of $Ca_3(PO_4)_2$ are actually obtained?

Strategy:

1. A balanced chemical equation is required to solve any stoichiometric problem.

$$Ca(OH)_2(s) + H_3PO_4(s) \rightarrow Ca_3(PO_4)_2(s) + ____$$

$Ca(OH)_2$ is a base and $H_3PO_4(s)$ is an acid, this is an acid-base reaction where $Ca_3(PO_4)_2$ is the salt result from the acid base reaction. The unspecified product in this reaction is water.

Complete the reaction and balance the equation:

$$\mathbf{3}Ca(OH)_2(s) + \mathbf{2}H_3PO_4(s) \rightarrow Ca_3(PO_4)_2(s) + \mathbf{6}H_2O(l)$$

2. Use the following steps to solve the stoichiometric problem.

Step a: Convert quantities of known substances into amount in mole

Reaction	3 $Ca(OH)_2$ +	2 H_3PO_4→	$Ca_3(PO_4)_2$ +	6 H_2O
Initial Amount	$\frac{9.8\ g}{(40.08+16.00*2+1.01*2)\ gmol^{-1}}$ = 0.13 mol	$\frac{9.8\ g}{98 gmol^{-1}}$ = 0.10 mol		

Step b: Since quantities of both reactants are known, limiting reactant (the one finishes first) need to be determined by dividing the amount of reactant with the stoichiometric coefficient in the balanced equation. The reactant that has lowest amount in per mole basis is the limiting reactant.

Reaction	3 $Ca(OH)_2$ +	2 H_3PO_4→	$Ca_3(PO_4)_2$ +	6 H_2O
Initial Amount	$\frac{9.8\ g}{(40.08+16.00*2+1.01*2)\ gmol^{-1}}$ = 0.13 mol	$\frac{9.8\ g}{98 gmol^{-1}}$ = 0.10 mol		
Divided by Coef.	$\frac{0.13\ mol}{3} = 0.043$	$\frac{0.10\ mol}{2} = 0.05$		
Identified as	Limiting Reactant	Reactant in Excess		

Step c: Use the mole ratio between the limiting reactant and unknown substance (i.e. $Ca_3(PO_4)_2$) to figure out the amount of unknown.

Reaction	3 $Ca(OH)_2$ +	2 H_3PO_4→	$Ca_3(PO_4)_2$ +	6 H_2O
Identified as	Limiting Reactant	Reactant in Excess		
Amount Reacted	0.13 mol		$\frac{n\ (Ca_3(PO_4)_2}{n(Ca(OH)_2)} = \frac{1}{3}$ $n(Ca_3(PO_4)_2) = \frac{1}{3} \times 0.13 = 0.043\ mol$	

Step d: Convert the amount of unknown into mass in gram.

Theoretical mass of $(Ca_3(PO_4)_2) = 0.043\ mol\ \times 310.3\ gmol^{-1}$

$= 13.446\ g$ (theoretical yield)

Step e: Find percentage yield of $Ca_3(PO_4)_2$:

$$\%\ yield = \frac{actual}{theoretical} \times 100\%$$

$$= \frac{2.5}{13.446} \times 100\%$$

$$= \mathbf{18.59}\%$$

Example 2.4:

Silicon steel is an alloy of the elements iron, carbon and silicon. An alloy sample was reacted with excess hydrochloric acid and the following reaction occurred:

$$Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$$

The carbon and silicon in the alloy did not react with the acid. If an alloy sample with a mass of 0.160 g produced 62.0 mL of hydrogen gas, measured at SLC, calculate:

a) the amount of hydrogen evolved in the reaction
b) the mass of iron that reacted to produce this amount of hydrogen
c) the percentage of iron in the alloy.

a $Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$

$$n(H_2) = \frac{0.062\ \text{L}}{24.5\ \text{L mol}^{-1}}$$

$= 2.5306 \times 10^{-3}$ mol

$= \mathbf{2.53 \times 10^{-3}\ mol}$

b $Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$

$$\frac{n(\text{Fe})}{n(\text{H}_2)} = \frac{1}{1}$$

$n(Fe) = n(H_2)$ from part **a**

$= 2.53 \times 10^{-3}$ mol

$m(Fe) = 2.53 \times 10^{-3}\ \text{mol} \times 55.8\ \text{g mol}^{-1}$

$= 0.14117$ g

$= \mathbf{0.141\ g}$

c $$\%\ (Fe) = \frac{m(\text{Fe})}{m(\text{alloy})} \times 100$$

$$= \frac{0.141}{0.160} \times 100$$

$= 88.12\%$

$= \mathbf{88.1\%}$

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Example 2.5:

Excessive intake of salt has been linked to increased blood pressure and heart disease. Many fast foods are high in salt.A 14.962 g sample of powdered chicken soup was mixed with approximately 100 mL of de-ionised water and stirred thoroughly. The mixture was filtered and the residue washed with more de-ionised water. The filtrate was made up to 250.0 mL in a volumetric flask. A 20.00 mL aliquot of this stock solution was pipetted into a conical flask. An excess of silver nitrate solution was then added.

The chloride ions present in the aliquot of stock solution were precipitated as silver chloride. The precipitate was filtered, dried overnight in an oven and weighed. The mass of silver chloride was 0.246 g.

Calculate the percentage by mass of sodium chloride in the powdered soup.

14.962 g chicken soup powder
+ 100 mL of water

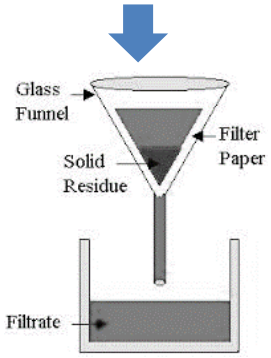


100 mL of chicken soup

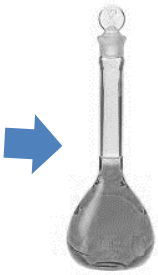
$NaCl + AgNO_3 \rightarrow NaNO_3 + AgCl$

Collect Precipitate, Dry Overnight and Weigh

m(AgCl) = 0.246 g

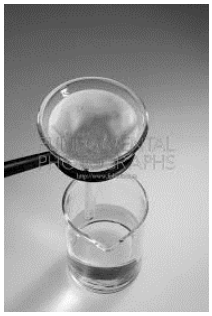


250 mL of stock solution



20 mL of stock solution + Excess of $AgNO_3$





$$m(AgCl)\ in\ 20\ mL\ of\ stock\ solution = 0.246\ g$$

$$n(AgCl)\ in\ 20\ mL\ of\ stock\ solution = \frac{0.246\ g}{143.32\ gmol^{-1}} = 0.00172\ mol$$

$$\frac{n(NaCl)}{n(AgCl)} = \frac{1}{1}$$

$$n(NaCl)\ in\ 20\ mL\ of\ stock\ solution = \ 0.00172\ mol$$

$$n(NaCl)\ in\ 250\ mL\ of\ stock\ solution = 0.00172 \times \frac{250\ mL}{20\ mL} = 0.0215\ mol$$

$$n(NaCl)\ in\ 100\ mL\ of\ chicken\ soup = n(NaCl)\ in\ 14.962\ g\ of\ sample$$

$$= n(NaCl) in\ 250\ mL\ of\ stock\ solution = 0.0215\ mol$$

$$m(NaCl) in\ original\ sample = 0.0215\ mol \times 58.5\ gmol^{-1} = 1.258\ g$$

$$\%\ of\ NaCl\ in\ sample = \frac{1.258}{14.962} \times 100\% = \mathbf{8.41}\ \%$$

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Self-Testing Questions

1. The combustion of Ethanol produces carbon dioxide and water.

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$$

If 100 g of ethanol completed combustion, calculate the volume of carbon dioxide (CO_2) produced, in L, at 100^0C and at 1.00 atm is:

A. 133
B. 35.5
C. 3587
D. 66.6

2. The initial reaction that occurs when a car airbag is triggered to inflate is

$$2NaN_3(s) \rightarrow 2Na(s) + 3N_2(g)$$

When 2.0 moles of NaN_3 reacts, the total volume, in L, of gas produced by this reaction, as measured at 32^0C and 100 kPa, is closest to

A. 7.98
B. 33.8
C. 50.7
D. 76.0

3. 100 mL of an 0.0100 M aqueous solution of calcium hydroxide will absorb carbon dioxide according to the equation:

$$Ca(OH)_2(aq) + 2CO_2(g) \rightarrow Ca(HCO_3)_2(aq)$$

The maximum volume of CO_2, in mL, at STP that could be absorbed by the solution is_____.

A. 22.4
B. 44.8
C. 224
D. 448

4. Carbon monoxide can be oxidised to carbon dioxide

$$2CO\ (g) + O_2(g) \rightarrow 2CO_2\ (g)$$

3 mole of CO and 2 mole of O_2 are mixed. When the reaction is complete there will be _____.

A. 4 mole of CO_2 produced.
B. 2 mole of CO_2 produced.
C. 1 mole of CO unreacted.
D. 0.5 mole of O_2 unreacted.

5. Hydrogen and chlorine react according to the equation

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$$

3 mole of H_2 and 2 mole of Cl_2 are placed in a vessel and sealed. When reaction is complete the vessel will contain_____.

A. 5 mole of HCl
B. 6 mole of HCl and 1 mole of Cl_2
C. 4 mole of HCl and 1 mole of Cl_2
D. 4 mole of HCl and 1 mole of H_2

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6. A reaction mixture containing 0.60 mol of aluminium (Al) and 1.20 mol of manganese dioxide (MnO_2) was prepared. The mixture was heated until one of the reactants has been completely consumed according to the equation:

$$4Al\ (s) + 3MnO_2(s) \rightarrow 3Mn(s) + 2Al_2O_3(s)$$

What is the amount of the excess reactant at the end of the reaction?

A. 0.20 mol Al
B. 0.40 mol Al
C. 0.30 mol MnO_2
D. 0.60 mol MnO_2

7. According to law of conservation mass, percent yield can never be greater than_____.
A. 100%
B. 75%
C. 50%
D. 25%

8. Ammonia is manufactured by the synthesis of nitrogen and hydrogen as shown in the following equation:

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

56.0 g of N_2 produced 34.0 g of NH_3. What is the percentage yield of ammonia?

A. 50%
B. 68%
C. 74%
D. 100%

9. Sulfur trioxide, SO_3 , is made from oxidation of SO_2 and is represented by the equation

$$2\ SO_2 + O_2 \rightarrow 2\ SO_3$$

What is the amount of SO_3 obtained if the reaction starts with 16.0 g of SO_2 and has a yield of 75%?

A. 20.0 g
B. 15.0 g
C. 12.0 g
D. 0.250 g

10. Copper reacts with dilute nitric acid to produce nitrogen monoxide according to the equation:

$$3Cu(s) + 8HNO_3(aq) \rightarrow 3Cu(NO_3)_2(aq) + 2NO(g) + 4H_2O(l)$$

When 6.75 g of Cu is added to 200.0 mL of 1.00 M HNO_3 and the reaction allowed to proceed to completion, 630 mL of NO is collected at STP. Which of the following is **closest** to the percentage of the NO produced in the reaction that was actually collected?

A. 27%
B. 40%
C. 56%
D. 60%

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11. Gravimetric analysis usually involves the following four procedures.

    I. Filtration
    II. Drying
    III. Precipitation
    IV. Weighing

    Arrange the procedure in the **correct** sequence.

    A. I, II, IV and III
    B. III, I, II and IV
    C. II, III, I and IV
    D. IV, III, I and II

12. The sodium chloride content of a 425 g can of baked beans was determined as follows. The Cl^- ions in 50.00 g of the baked beans were precipitated as silver chloride, AgCl. The mass of the precipitate was 1.051 g. The amount of NaCl in the original can was closest to:

    A. 0.00733 mol
    B. 0.0180 mol
    C. 0.0623 mol
    D. 0.153 mol

13. The amount of sulfate in a fertilizer was determined by gravimetric analysis. By adding excess $BaCl_2$ solution, highly insoluble $BaSO_4$ was produced, which was filtered out, washed and dried. If the manufacturer's label states that the concentration of sulfur in the fertiliser is 11 %(w/w), what should be the approximate dry mass of $BaSO_4$ precipitate produced if the mass of the fertiliser sample under analysis is 1.000 g?

    A. 0.27 g
    B. 0.80 g
    C. 2.7 g
    D. 8.0

Question 14 and 15 are related to the following information.

The percentage by mass of sodium ions in the form of sodium chloride in a particular brand of dry biscuit was determined using gravimetric analysis. A 3.65 g sample of the biscuits was ground into powder, dissolved in water and filtered. Excess silver nitrate was added to the remaining solution to precipitate the chloride ions as silver chloride. The precipitate was collected, washed and dried. Its mass was found to be 0.241 g.

14. The percentage by mass of **sodium ions (Na^+)** in the biscuit is closest to:

    A. 0.0387 %
    B. 1.06%
    C. 1.47%
    D. 2.69%

15. The calculated percentage by mass of sodium chloride is lower than stated on the label on the packaging. This may be due to:

    A. Incomplete precipitation of the chloride ions.
    B. Inadequate washing of the precipitate.
    C. The precipitate not being dried to constant mass.
    D. Co-precipitation of another, unknown anion with the silver ions.

Answer Keys

1 A **2** D **3** B **4** D **5** D **6** C **7** A **8** A **9** B **10** C **11** B **12** C **13** B **14** B **15** A

Past Exam Questions

1. Coral reefs are produced by living organisms and predominantly made up of calcium carbonate. One method of determining the proportion of calcium carbonate in a coral is to dissolve a known mass of the coral in excess acid and measure the volume of carbon dioxide formed.

(a) Complete the equation below for the reaction between calcium carbonate and hydrochloric acid: **[1m]**

$$\text{.....}CaCO_3(s) + \text{.....}HCl(aq) \rightarrow \text{.....}CaCl_2(aq) + \text{.....}H_2O(l) + \text{.....}CO_2(g)$$

In such an experiment, 1.13 g of coral was dissolved in excess hydrochloric acid in a conical flask. When the reaction was complete, 224 mL of carbon dioxide gas had been collected at room temperature and pressure.

(b) Calculate the amount, in mole, of carbon dioxide collected in this experiment. **[2m]**
[Given the molar volume of any gas at room temperature and pressure is 24.5 L]

(c) Use the data and balanced equation given, calculate the <u>mass of calcium carbonate</u> in the sample and the <u>percentage by mass</u> of calcium carbonate in the coral. **[4m]**

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2. Airbags in cars are designed to expand in less than 40 milliseconds when a collision occurs, to reduce the risk of severe injury. When ignited with an electric current, a mixture of sodium azide (NaN_3)and sodium nitrate ($NaNO_3$) will react explosively. The reaction can be represented by the following equation:

$$NaNO_3(s) + 5NaN_3(s) \rightarrow 3Na_2O(s) + 8N_2(g)$$

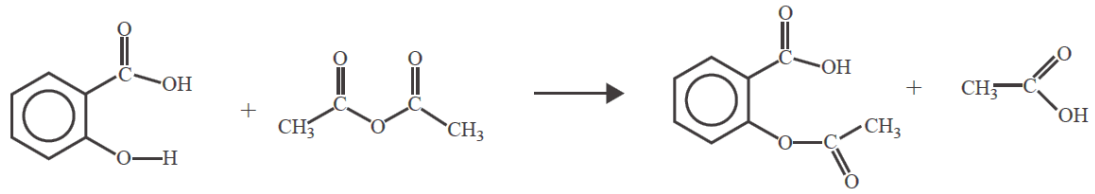
(a) An airbag was produced containing 205.0 g of sodium nitrate and 750.0 g of sodium azide. Determine the limiting reactant in this reaction. **[4m]**

(b) Calculate the volume of the nitrogen gas will inflate in the airbag at STP. **[2m]**

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3. A sample of aspirin was prepared by reacting 2.20 g of salicylic acid with 4.20 mL of ethanoic anhydride in a conical flask. After heating for 20 minutes the reaction mixture was cooled and white crystals precipitated. The crystals were then collected, dried to constant mass and weighed. The mass of product obtained was 2.25 g.

   The equation for the reaction is:



salicylic acid (s) ethanoic anhydride (l) aspirin (s)

Use the following data to answer the questions below.

	Molar mass (g mol^{-1})	**Density (g mL^{-1})**
Salicylic acid	138	
Ethanoic anhydride	102	1.08
Aspirin	180	

(a) Calculate the initial amount, in moles, of salicylic acid used in this preparation. **[2m]**

(b) Calculate the initial amount, in moles, of ethanoic anhydride used.**[3m]**

(c) Calculate the maximum mass of aspirin that can theoretically be produced from these reagents.**[3m]**

(d) Determine the percentage yield in this preparation. **[2m]**

4. Urea, $CO(NH_2)_2$, a commonly used fertiliser, is produced as crystals by the reaction of ammonia with carbon dioxide as shown in the following equation:

$$2NH_3(g) + CO_2(g) \rightarrow CO(NH_2)_2\ (s) + H_2O(g)$$

A reaction vessel designed for the synthesis of urea is operated at 200^0C and 148 atm. It has a total volume of 5000 L, and ammonia (NH_3) and carbon dioxide (CO_2) are fed into it in batches so that ammonia occupies 62.0% of the volume and carbon dioxide the remainder.

Note: 1 atm = 101 kPa; T (Kelvin) = T^0C + 273; gas constant, R = 8.31 J K^{-1} mol^{-1}

(a) Calculate the amount of ammonia (NH_3) and carbon dioxide (CO_2) used in the above reaction. **[4m]**

(b) Determine which reactant (NH_3 or CO_2) is the limiting reactant. Show all your workings. **[3m]**

(c) Calculate the maximum mass of urea, in kg, that can be produced from the above reaction. **[3m]**

(d) Calculate the amount of excess reactant, in mol, after the reaction is complete. **[2m]**

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5. A blast furnace is a large furnace operated at very high temperatures to convert iron (III) oxide (in iron ore) to iron using carbon monoxide, which reaction can be represented as follow.

$$____ Fe_2O_3(s) + ____CO(g) \rightarrow _____ Fe(s) + _____CO_2(g)$$

(a) Balance the above reaction equation. **[1m]**

(b) Iron ore that contains 0.965 tonne of iron(III) oxide, Fe_2O_3, is fed into the blast furnace with 2.70×10^6 L of carbon monoxide at 1.12 atm pressure and 1986^0C.
Note: 1 tonne= 1000 kg, 1 atm = 101 kPa; gas constant, R = 8.314; T (K) = ^{0}C + 273

(i) Determine the limiting reactant for this reaction.**[4m]**

(ii) What mass of iron is theoretically produced in this reaction?**[3m]**

(iii) Calculate the mass of the reactant in excess.**[3m]**

(c) If 5.56×10^{-1} tonne of iron is actually produced, what is the overall percentage yield of the process?**[2m]**